



Guidelines

Renal Mass and Localized Renal Cancer: Evaluation, Management, and Follow Up (2021)

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[RCC Follow-up Algorithm](#) [pdf]

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Executive Summary

Purpose

This AUA Guidelines focuses on the evaluation and management of clinically localized sporadic renal masses suspicious for renal cell carcinoma (RCC) in adults, including solid enhancing renal tumors and Bosniak 3 and 4 complex cystic renal masses. Some patients with clinically localized renal masses may present with findings suggesting aggressive tumor biology or may be upstaged on exploration or final pathology. Management considerations pertinent to the urologist in such patients will also be discussed. The follow-up of renal cancer patients after intervention is also addressed, including recommendations for periodic clinical follow-up and abdominal and chest imaging. Practice patterns regarding such tumors vary considerably, and the literature regarding evaluation, management, and surveillance has been rapidly evolving.

Notable examples include controversies about the role of renal mass biopsy (RMB) and concerns regarding overutilization of radical nephrectomy (RN). Please also refer to the associated Renal Mass and Localized Renal Cancer Treatment and Follow-up after Intervention algorithms.

Methodology

The systematic review utilized in the creation of this guideline was completed in part through the Agency for Healthcare Research and Quality (AHRQ) and through additional supplementation that further addressed additional key questions and more recently published literature. A research librarian experienced in conducting literature searches for comparative effectiveness reviews searched in MEDLINE®, Embase®, the Cochrane Library, the Database of Abstracts of Reviews of Effects, the Health Technology Assessment Database, and the UK National Health Service Economic Evaluation database to capture both published and gray literature published from January 1, 1997 through May 1, 2015. A supplemental search was conducted adding additional literature published through August 2015, and a final update search was conducted through July 2016.

A systematic review was conducted in 2013 to identify published articles relevant to key questions specified by the Panel related to kidney neoplasms and their follow-up (imaging, renal function, markers, biopsy, and prognosis). This search covered English-language articles published between January 1999 and 2011. An updated query was later conducted to include studies published through August 2012.

In January of 2021, the Renal Mass and Localized Renal Cancer guideline underwent an additional amendment based on a current literature search. This literature search retrieved additional studies published between July 2016 to October 2020 using the same Key Questions and search criteria from the Renal Mass and Localized Renal Cancer guideline. Nineteen studies were identified from this search to provide data relevant to the management and treatment of Renal Mass. In addition, the Follow-Up for Clinically Localized Renal Neoplasms guideline published in 2013 was merged with the Renal Mass and Localized Renal Cancer guideline. Although the systematic search for follow-up interventions was not updated to 2020, the panel members conducted a comprehensive review of all evidence published since the original guideline. The language of many statements has been refined for clarity. For all evidence-based statements, supporting studies were identified only in the original systematic review and the evidence strength was not altered.

When sufficient evidence existed, the body of evidence for a particular treatment was assigned a strength rating of A (high), B (moderate) or C (low) for support of Strong, Moderate, or Conditional Recommendations. In the absence of sufficient evidence, additional information is provided as Clinical Principles and Expert Opinions.

Initial Evaluation and Diagnosis

Evaluation

1. In patients with a solid or complex cystic renal mass, clinicians should obtain high quality, multiphase, cross-sectional abdominal imaging to optimally characterize and clinically stage the renal mass. Characterization of the renal mass should include assessment of tumor complexity, degree of contrast enhancement (where applicable), and presence or absence of fat. (Clinical Principle)
2. In patients with suspected renal malignancy, clinicians should obtain a comprehensive metabolic panel, complete blood count, and urinalysis. Metastatic evaluation should include chest imaging to evaluate for possible thoracic metastases. (Clinical Principle)
3. For patients with a solid or Bosniak 3/4 complex cystic renal mass, clinicians should assign chronic kidney disease (CKD) stage based on glomerular filtration rate (GFR) and degree of proteinuria. (Expert Opinion)

Counseling

4. In patients with a solid or Bosniak 3/4 complex cystic renal mass, a urologist should lead the counseling process and should consider all management strategies. A multidisciplinary team should be included when necessary. (Expert Opinion)
5. Clinicians should provide counseling that includes current perspectives about tumor biology and a patient-specific risk assessment inclusive of sex, tumor size/complexity, histology (when obtained), and imaging characteristics. For cT1a tumors, the low oncologic risk of many small renal masses should be reviewed. (Clinical Principle)
6. During counseling of patients with a solid or Bosniak 3/4 complex cystic renal mass, clinicians must review the most common and serious urologic and non-urologic morbidities of each treatment pathway and the importance of patient age, comorbidities/frailty, and life expectancy. (Clinical Principle)

7. Clinicians should review the importance of renal functional recovery related to renal mass management, including the risks of progressive CKD, potential short- or long-term need for renal replacement therapy, and long-term overall survival considerations. (Clinical Principle)
8. Clinicians should consider referral to nephrology in patients with a high risk of CKD progression, including those with estimated glomerular filtration rate (eGFR) less than 45 mL/min/1.73m², confirmed proteinuria, diabetics with preexisting CKD, or whenever eGFR is expected to be less than 30 mL/min/1.73m² after intervention. (Expert Opinion)
9. Clinicians should recommend genetic counseling for any of the following: all patients ≤ 46 years of age with renal malignancy, those with multifocal or bilateral renal masses, or whenever 1) the personal or family history suggests a familial renal neoplastic syndrome; 2) there is a first- or second-degree relative with a history of renal malignancy or a known clinical or genetic diagnosis of a familial renal neoplastic syndrome (even if kidney cancer has not been observed); or 3) the patient's pathology demonstrates histologic findings suggestive of such a syndrome. (Expert Opinion)

Renal Mass Biopsy (RMB)

10. When considering the utility of RMB, patients should be counseled regarding rationale, positive and negative predictive values, potential risks and non-diagnostic rates of RMB. (Moderate Recommendation; Evidence Level: Grade C)
11. Clinicians should consider RMB when a mass is suspected to be hematologic, metastatic, inflammatory, or infectious. (Clinical Principle)
12. In the setting of a solid renal mass, RMB should be obtained on a utility-based approach whenever it may influence management. RMB is not required for 1) young or healthy patients who are unwilling to accept the uncertainties associated with RMB; or 2) older or frail patients who will be managed conservatively independent of RMB findings. (Expert Opinion)
13. For patients with a solid renal mass who elect RMB, multiple core biopsies should be performed and are preferred over fine needle aspiration (FNA). (Moderate Recommendation; Evidence Level: Grade C)

Management

Partial nephrectomy (PN) and nephron-sparing approaches

14. Clinicians should prioritize PN for the management of the cT1a renal mass when intervention is indicated. In this setting, PN minimizes the risk of CKD or CKD progression and is associated with favorable oncologic outcomes, including excellent local control. (Moderate Recommendation; Evidence Level: Grade B)
15. Clinicians should prioritize nephron-sparing approaches for patients with solid or Bosniak 3/4 complex cystic renal masses and an anatomic or functionally solitary kidney, bilateral tumors, known familial RCC, preexisting CKD, or proteinuria. (Moderate Recommendation; Evidence Level: Grade C)
16. Nephron-sparing approaches should be considered for patients with solid or Bosniak 3/4 complex cystic renal masses who are young, have multifocal masses, or comorbidities that are likely to impact renal function in the future, including but not limited to moderate to severe hypertension, diabetes mellitus, recurrent urolithiasis, or morbid obesity. (Moderate Recommendation; Evidence Level: Grade C)
17. In patients who elect PN, clinicians should prioritize preservation of renal function by optimizing nephron mass preservation and avoiding prolonged warm ischemia. (Expert Opinion)
18. For patients undergoing PN, clinicians should prioritize negative surgical margins. The extent of normal parenchyma removed should be determined by surgeon discretion taking into account the clinical situation and tumor characteristics, including growth pattern, and interface with normal tissue. Tumor enucleation should be considered in patients with familial RCC, multifocal disease, or severe CKD to optimize parenchymal mass preservation. (Expert Opinion)

Radical Nephrectomy (RN)

19. Clinicians should consider RN for patients with a solid or Bosniak 3/4 complex cystic renal mass whenever increased oncologic potential is suggested by tumor size, RMB (if obtained), and/or imaging. (Moderate Recommendation; Evidence Level: Grade B) In this setting, RN is preferred if *all* of the following criteria are met: 1) high tumor complexity and PN would be challenging even in experienced hands; 2) no preexisting CKD or proteinuria; and 3) normal

contralateral kidney and new baseline eGFR will likely be greater than 45 mL/min/1.73m² even if RN is performed. If all of these criteria are not met, PN should be considered unless there are overriding concerns about the safety or oncologic efficacy of PN. (Expert Opinion)

Surgical Principles

20. For patients who are undergoing surgical excision of a renal mass with clinically concerning regional lymphadenopathy, clinicians should perform a lymph node dissection including all clinically positive nodes for staging purposes. (Expert Opinion)
21. For patients who are undergoing surgical excision of a renal mass, clinicians should perform adrenalectomy if imaging and/or intraoperative findings suggest metastasis or direct invasion of the adrenal gland. (Clinical Principle)
22. In patients undergoing surgical excision of a renal mass, a minimally invasive approach should be considered when it would not compromise oncologic, functional, and perioperative outcomes. (Expert Opinion)

Other Considerations

23. Pathologic evaluation of the adjacent renal parenchyma should be performed and recorded after PN or RN to assess for possible intrinsic renal disease, particularly for patients with CKD or risk factors for developing CKD. (Clinical Principle)
24. Clinicians should consider referral to medical oncology whenever there is concern for potential clinical metastasis or incompletely resected disease (macroscopic positive margin or gross residual disease). Patients with high-risk or locally advanced, fully resected renal cancers should be counselled about the risks/benefits of adjuvant therapy and encouraged to participate in adjuvant clinical trials, facilitated by medical oncology consultation when needed. (Clinical Principle)

Thermal Ablation (TA)

25. Clinicians should consider TA as an alternate approach for the management of cT1a solid renal masses <3 cm in size. For patients who elect TA, a percutaneous technique is preferred over a surgical approach whenever feasible to minimize morbidity. (Moderate Recommendation; Evidence Level: Grade C)
26. Both radiofrequency ablation (RFA) and cryoablation may be offered as options for patients who elect TA. (Conditional Recommendation; Evidence Level: Grade C)
27. A RMB should be performed prior to (preferred) or at the time of ablation to provide pathologic diagnosis and guide subsequent surveillance. (Expert Opinion)
28. Counseling about TA should include information regarding an increased likelihood of tumor persistence or local recurrence after primary TA relative to surgical excision, which may be addressed with repeat ablation if further intervention is elected. (Strong Recommendation; Evidence Level: Grade B)

Active Surveillance (AS)

29. For patients with a solid renal mass <2cm, or those that are complex but predominantly cystic, clinicians may elect AS with potential for delayed intervention for initial management. (Conditional Recommendation; Evidence Level: Grade C)
30. For patients with a solid or Bosniak 3/4 complex cystic renal mass, clinicians should prioritize AS/expectant management when the anticipated risk of intervention or competing risks of death outweigh the potential oncologic benefits of active treatment. In asymptomatic patients, the panel recommends periodic clinical surveillance and/or imaging based on shared decision making. (Clinical Principle)
31. For patients with a solid or Bosniak 3/4 complex cystic renal mass in whom the risk/benefit analysis for treatment is equivocal and who prefer AS, clinicians should consider RMB (if the mass is solid or has solid components) for further oncologic risk stratification. Repeat cross-sectional imaging should be obtained approximately 3-6 months later to assess for interval growth. Periodic clinical/imaging surveillance can then be based on growth rate and shared decision-making with intervention recommended if substantial interval growth is observed or if other clinical/imaging findings suggest that the risk/benefit analysis is no longer equivocal or favorable for continued AS. (Expert Opinion)
32. For patients with a solid or Bosniak 3/4 complex cystic renal mass in whom the anticipated oncologic benefits of intervention outweigh the risks of treatment and competing risks of death, clinicians should recommend intervention.

AS with potential for delayed intervention may be pursued only if the patient understands and is willing to accept the associated oncologic risks. In this setting, clinicians should encourage RMB (if the mass is predominantly solid) for additional risk stratification. If the patient continues to prefer AS, close clinical and cross-sectional imaging surveillance with periodic reassessment and counseling should be recommended. (Moderate Recommendation; Evidence Level: Grade C)

Follow-Up after Intervention

General Priniciples

33. Clinicians coordinating follow-up for patients who have undergone intervention for a renal mass should discuss the implications of stage, grade, and histology including the risks of recurrence and possible sequelae of treatment. Patients with pathologically-proven benign renal masses should undergo occasional clinical evaluation and laboratory testing for sequelae of treatment but most do not require routine periodic imaging. (Expert Opinion)
34. Patients with treated malignant renal masses should undergo periodic medical history, physical examination, laboratory studies, and imaging directed at detecting signs and symptoms of metastatic spread and/or local recurrence as well as evaluation for possible sequelae of treatment. (Clinical Principle)
35. Patients with treated malignant renal masses should have periodic laboratory testing including serum creatinine, eGFR, and urinalysis. Other laboratory evaluations (e.g., complete blood count, lactate dehydrogenase, liver function tests, alkaline phosphatase and calcium level) may be obtained at the discretion of the clinician or if advanced disease is suspected. (Expert Opinion)
36. Patients undergoing follow-up for treated renal masses with progressive renal insufficiency or proteinuria should be referred to nephrology. (Expert Opinion)
37. Patients undergoing follow-up for treated malignant renal masses should only undergo bone scan if one or more of the following is present: clinical symptoms such as bone pain, elevated alkaline phosphatase, or radiographic findings suggestive of a bony neoplasm. (Moderate Recommendation; Evidence Level: Grade C)
38. Patients undergoing follow-up for treated malignant renal masses with acute neurological signs or symptoms should undergo prompt magnetic resonance imaging (MRI) or computed tomography (CT) scanning of the brain and/or spine. (Strong Recommendation; Evidence Level: Grade A)
39. For patients undergoing follow-up for treated malignant renal masses, additional site-specific imaging can be ordered as warranted by clinical symptoms suggestive of recurrence or metastatic spread. Positron emission tomography (PET) scan should not be obtained routinely but may be considered selectively. (Moderate Recommendation; Evidence Level: Grade C)
40. Patients with findings suggestive of metastatic renal malignancy should be evaluated to define the extent of disease and referred to medical oncology. Surgical resection or ablative therapies should be considered in select patients with isolated or oligo-metastatic disease. (Expert Opinion)
41. Patients with findings suggesting a new renal primary or local recurrence of renal malignancy should undergo metastatic evaluation including chest and abdominal imaging. If the new primary or recurrence is isolated to the ipsilateral kidney and/or retroperitoneum, a urologist should be involved in the decision-making process, and surgical resection or ablative therapies may be considered. (Expert Opinion)

Follow-up after Surgery

42. 42. Clinicians should classify patients who have been managed with surgery (PN or RN) for a malignant renal mass into one of the following risk groups for follow-up:

Low Risk (LR):	pT1 and Grade 1/2
Intermediate Risk (IR):	pT1 and Grade 3/4, or pT2 any Grade
High Risk (HR):	pT3 any Grade

Very High Risk (VHR):	pT4 or pN1, or sarcomatoid/rhabdoid dedifferentiation, or macroscopic positive margin
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If final microscopic surgical margins are positive for cancer, the risk category should be considered at least one level higher, and increased clinical vigilance should be exercised. (Expert Opinion)

43. Patients managed with surgery (PN or RN) for a renal malignancy should undergo abdominal imaging according to Table 1, with CT or MRI pre- and post-intravenous contrast preferred. (Moderate Recommendation; Evidence Strength: Grade C). After 2 years, abdominal ultrasound (US) alternating with cross-sectional imaging may be considered in the LR and IR groups at physician discretion. After 5 years, informed/shared decision-making should dictate further abdominal imaging. (Expert Opinion)
44. Patients managed with surgery (PN or RN) for a renal malignancy should undergo chest imaging (chest x-ray [CXR] for LR and IR; CT chest preferred for HR and VHR) according to Table 1. (Moderate Recommendation; Evidence Strength: Grade C). After 5 years, informed/shared decision-making discussion should dictate further chest imaging and CXR may be utilized instead of chest CT for HR and VHR (Expert Opinion)

Table 1: Recommended follow-up schedule after surgery for renal cancer (in months)*												
Risk	3	6	9	12	18	24	30	36	48	60	72-84	96-120
LR				X		X			X	X	X	X
IR		X		X		X		X	X	X	X	X
HR		X		X	X	X	X	X	X	X	X	X
VHR	X	X	X	X	X	X	X	X	X	X	X	X
*Follow-up timeline is approximate and allows flexibility to accommodate reasonable patient, caregiver, and institutional needs. Each follow-up visit should include relevant history, physical examination, laboratory testing, and abdominal and chest imaging. Overall, 30% of renal cancer recurrences after surgery are diagnosed beyond 60 months. ¹ Informed/shared decision-making should guide surveillance decisions beyond 60 months.												

Follow-up after TA

45. Patients undergoing ablative procedures with biopsy that confirmed malignancy or was non-diagnostic should undergo pre- and post-contrast cross-sectional abdominal imaging within 6 months (if not contraindicated). Subsequent follow-up should be according to the recommendations for the IR postoperative protocol (Table 1). (Expert Opinion)

Introduction

Purpose

This AUA Guidelines focuses on the evaluation and management of clinically localized sporadic renal masses suspicious for RCC in adults, including solid enhancing renal tumors and Bosniak 3 and 4 complex cystic renal masses. Some patients with clinically localized renal masses may present with findings suggesting aggressive tumor biology or may be upstaged on exploration or final pathology. Management considerations pertinent to the urologist in such patients will also be discussed. The follow-up of renal cancer patients after intervention is also addressed including recommendations for periodic clinical follow-up and abdominal and chest imaging. Practice patterns regarding such tumors vary considerably and the literature regarding evaluation, management, and surveillance has been rapidly evolving. Notable examples include controversies about the role of RMB and concerns about overutilization of RN.

Methodology

Systematic Review RM

The systematic review utilized in the creation of this guideline was completed in part through the AHRQ and through additional supplementation that further addressed additional key questions and more recently published literature. A research librarian experienced in conducting literature searches for comparative effectiveness reviews searched in MEDLINE®, Embase®, the Cochrane Library, the Database of Abstracts of Reviews of Effects, the Health Technology Assessment Database, and the UK National Health Service Economic Evaluation database to capture both published and gray literature published from January 1, 1997 through May 1, 2015. A supplemental search was conducted adding additional literature published through August 2015, and a final update search was conducted through July 2016.

Systematic Review FRC

A systematic review was conducted to identify published articles relevant to key questions specified by the Panel related to kidney neoplasms and their follow-up (imaging, renal function, markers, biopsy, and prognosis). This search covered articles in English published between January 1999 and 2011. An updated query was later conducted to include studies published through August 2012. Study designs consisting of clinical trials (randomized or not), observational studies (cohort, case-control, case series) and systematic reviews were included. All other study types were excluded. Studies with full-text publication available were included, but studies in abstract form only were excluded.

Combination of Guidelines

In January of 2021, the Renal Mass and Localized Renal Cancer guideline underwent an additional amendment based on current literature. The updated literature search retrieved additional studies published between July 2016 to October 2020 using the same search strategy from the Renal Mass and Localized Renal Cancer guideline. Following study selection using the original PICO criteria, as reporting data relevant to the management and treatment of Renal Mass. In addition, the Follow-Up for Clinically Localized Renal Neoplasms guideline published in 2013 was merged with the Renal Mass and Localized Renal Cancer guideline. Although the systematic search for follow-up interventions was not updated to 2020, the panel members conducted a comprehensive review of all evidence published since the original guideline. The language of many statements has been refined for clarity. For all evidence-based statements, supporting studies were identified only in the original systematic review and the evidence strength was not altered.

Assessment of Risk-of-Bias of Individual Studies

Citations identified by the systematic search were screened independently by two reviewers using predefined PICO criteria. One reviewer completed data abstraction and a second reviewer checked abstraction for accuracy. Two reviewers independently assessed risk of bias for individual studies. The Cochrane Collaboration’s tool was used for assessing the risk of bias of randomized controlled trials (RCTs).² For nonrandomized studies of treatment interventions, the reviewers used the Risk of Bias in Non-Randomized Studies – of Intervention (ROBINS-I). For diagnostic studies, we used the quality assessment tool for diagnostic accuracy studies (QUADAS -2).^{3,4} Differences between reviewers were resolved through consensus.

Determination of Evidence Strength

The categorization of evidence strength is conceptually distinct from the quality of individual studies. Evidence strength refers to the body of evidence available for a particular question and includes not only individual study quality but consideration of study design, consistency of findings across studies, adequacy of sample sizes, and generalizability of samples, settings, and treatments for the purposes of the guideline. The AUA categorizes body of evidence strength as Grade A (well-conducted and highly-generalizable RCTs or exceptionally strong observational studies with consistent findings), Grade B (RCTs with some weaknesses of procedure or generalizability or moderately strong observational studies with consistent findings), or Grade C (RCTs with serious deficiencies of procedure or generalizability or extremely small sample sizes or observational studies that are inconsistent, have small sample sizes, or have other problems that potentially confound interpretation of data). By definition, Grade A evidence is evidence about which the Panel has a high level of certainty, Grade B evidence is evidence about which the Panel has a moderate level of certainty, and Grade C evidence is evidence about which the Panel has a low level of certainty.⁵

AUA Nomenclature: Linking Statement Type to Evidence Strength

The AUA nomenclature system explicitly links statement type to body of evidence strength, level of certainty, magnitude of benefit or risk/burdens, and the Panel’s judgment regarding the balance between benefits and risks/burdens (Table 2). **Strong Recommendations** are directive statements that an action should (benefits outweigh risks/burdens) or should not

(risks/burdens outweigh benefits) be undertaken because net benefit or net harm is substantial. **Moderate Recommendations** are directive statements that an action should (benefits outweigh risks/burdens) or should not (risks/burdens outweigh benefits) be undertaken because net benefit or net harm is moderate. **Conditional Recommendations** are non-directive statements used when the evidence indicates that there is no apparent net benefit or harm or when the balance between benefits and risks/burden is unclear. All three statement types may be supported by any body of evidence strength grade. Body of evidence strength Grade A in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances and that future research is *unlikely to change confidence*. Body of evidence strength Grade B in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances but that better evidence *could change confidence*. Body of evidence strength Grade C in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances but that better evidence *is likely to change confidence*. Body of evidence strength Grade C is only rarely used in support of a Strong Recommendation. Conditional Recommendations can also be supported by any evidence strength. When body of evidence strength is Grade A, the statement indicates that benefits and risks/burdens appear balanced, the best action depends on patient circumstances, and future research is *unlikely to change confidence*. When body of evidence strength Grade B is used, benefits and risks/burdens appear balanced, the best action also depends on individual patient circumstances and better evidence could change confidence. When body of evidence strength Grade C is used, there is uncertainty regarding the balance between benefits and risks/burdens, alternative strategies may be equally reasonable, and better evidence is *likely to change confidence*.

Where gaps in the evidence existed, the Panel provides guidance in the form of *Clinical Principles* or *Expert Opinion* with consensus achieved using a modified Delphi technique if differences of opinion emerged.⁶ A *Clinical Principle* is a statement about a component of clinical care that is widely agreed upon by urologists or other clinicians for which there may or may not be evidence in the medical literature. *Expert Opinion* refers to a statement, achieved by consensus of the Panel, that is based on members' clinical training, experience, knowledge, and judgment for which there is no evidence.

Table 2: AUA Nomenclature Linking Statement Type to Level of Certainty, Magnitude, of Benefits or Risk/Burden, and Body of Evidence Strength			
	Evidence Strength A (High Certainty)	Evidence Strength B (Moderate Certainty)	Evidence Strength C (Low Certainty)
Strong Recommendation (Net benefit or harm substantial)	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) is substantial Applies to most patients in most circumstances and future research is unlikely to change confidence	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) is substantial Applies to most patients in most circumstances and future research could change confidence	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) appears substantial Applies to most patients in most circumstances and future research is likely to change confidence (rarely used to support a Strong Recommendation)
Moderate Recommendation (Net benefit or harm moderate)	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) is moderate Applies to most patients in most circumstances and future research is unlikely to change confidence	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) is moderate Applies to most patients in most circumstances and future research could change confidence	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) appears moderate Applies to most patients in most circumstances and future research is likely to change confidence

Table 2: AUA Nomenclature Linking Statement Type to Level of Certainty, Magnitude, of Benefits or Risk/Burden, and Body of Evidence Strength			
Conditional Recommendation (No apparent net benefit or harm)	Benefits = Risks/Burdens (or vice versa) Best action depends on individual patient circumstances Future research unlikely to change confidence	Benefits = Risks/Burdens (or vice versa) Best action appears to depend on individual patient circumstances Better evidence could change confidence	Balance between Benefits and Risks/Burdens unclear Alternative strategies may be equally reasonable Better evidence likely to change confidence
Clinical Principle	A statement about a component of clinical care that is widely agreed upon by urologists or other clinicians for which there may or may not be evidence in the medical literature		
Expert Opinion	A statement, achieved by consensus of the Panel, that is based on members' clinical training, experience, knowledge, and judgment for which there may or may not be evidence in the medical literature		

Process

The Renal Mass and Localized Renal Cancer Panel was created in 2014 by the American Urological Association Education and Research, Inc. (AUA). The Practice Guidelines Committee (PGC) of the AUA selected the Panel Chair who in turn appointed the Vice Chair. In a collaborative process, additional Panel members, including additional members of the College of American Pathologists (CAP), Society of Urologic Oncology (SUO), American College of Radiology (ACR), American Society of Nephrology (ASN), Endourological Society, and Society of Interventional Radiology (SIR) with specific expertise in this area, were then nominated and approved by the PGC. The AUA conducted a thorough peer review process. The draft guidelines document was distributed to 124 peer reviewers, 54 of which submitted comments. The Panel reviewed and discussed all submitted comments and revised the draft as needed. Once finalized, the guideline was submitted for approval to the PGC and Science and Quality Council. Then it was submitted to the AUA, CAP, SUO, ACR, ASN, Endourological Society, and SIR Board of Directors for final approval. Panel members received no remuneration for their work.

A Renal Mass amendment panel consisting of five members was created in August 2020 to conduct an update to the Renal Mass and Localized Renal Cancer guideline. They were also tasked with integrating the Follow-Up for Clinically Localized Renal Neoplasms guideline from 2013 into the Renal Mass and Localized Renal Cancer guideline from 2017 to create one cohesive document on management and follow-up on renal mass. The panel consisted of members from both the Follow-Up for Clinically Localized Renal Neoplasms guideline and the Renal Mass and Localized Renal Cancer guideline with an additional new member who has not previously served on one of these panels. The AUA conducted a thorough peer review process. The draft guidelines document was distributed to 75 peer reviewers, 21 of which submitted comments. The Panel reviewed and discussed all submitted comments and revised the draft as needed. Once finalized, the guideline was submitted for approval to the PGC and Science and Quality Council and the AUA Board of Directors for final approval. Panel members received no remuneration for their work.

Background

Renal masses are a biologically heterogeneous group of tumors ranging from benign masses to cancers that can be indolent or aggressive.^{7,8} The true incidence of renal masses (including benign masses) is unknown. However, benign masses comprise approximately 15-20 percent of surgically resected tumors < 4 cm and allow estimations of benign incidence based on kidney cancer statistics.^{7,9,10} The vast majority (greater than 90%) of kidney cancers in the United States are renal cortical tumors known as RCC.

Epidemiology: United States

It is estimated there will be over 73,000 new cases of kidney cancer in the United States in 2020.^{11,12} The incidence of kidney cancer has been increasing steadily since the 1970’s in part due to more prevalent use of axial imaging (CT and MRI).¹³ In the United States, over the past decade, the incidence of kidney cancer continues to increase but at a much

smaller increment, approximately 1% per year. The greatest increase in incidence has been in small, clinically localized renal masses which now represent at least 40 percent of incident tumors.^{14,15}

The overall survival rate for all stages of renal cancer is approximately 74%, leaving an estimated 400,000 kidney cancer survivors in the United States as of 2013.¹¹ However, approximately 14,800 men and women will die of kidney cancer in 2020.¹⁰ The mortality from kidney cancer has been steadily decreasing, approximately 1% per year, since 2004.^{16,17} Reasons for this decrease are multifactorial.

Kidney cancer is more common in men than women, and more common in African Americans, American Indian and Alaska Native populations than Caucasians.¹⁸ The median age at diagnosis is 64 years old, although kidney cancer can present at any age.¹⁹

Epidemiology: Global and International Considerations

Over 300,000 men and women are diagnosed with kidney cancer around the world each year and approximately 150,000 patients will die of disease.²⁰ The incidence of kidney cancer varies dramatically around the world with the developed countries having the highest rates.²¹ Incidence rates have increased in both sexes and are most notable in the elderly population (greater than 75 years of age). Mortality rates have been stable in most countries but have been decreasing by 1 to 3 percent in Western and Northern Europe, the United States, and Australia. The improved mortality globally and in the US is attributed to decreased smoking rates, improved therapies, and access to medical care. The decrease in mortality has been faster in women than in men and overall mortality rates remain higher in men than women.

Etiology

There are a number of established and putative risk factors for RCC. Smoking is a well-established risk factor, accounting for 20 percent of incident cases and increasing the risk of RCC by 50 percent in men and 20 percent in women.^{22,23} Obesity is associated with 30% of incident cases of RCC and each 5 kg/m² increase in body mass index increases the risk of RCC by 24 percent in men and 34 percent in women.²³⁻²⁵ Interestingly, an “obesity paradox” exists in kidney cancer – where obese patients are more likely to develop RCC, but these tumors are more likely to be low-grade, early stage tumors.²⁵⁻²⁷ Hypertension is also associated with increased risk of RCC.^{23,28,29} The role of CKD as a risk factor is controversial; however, patients on maintenance dialysis are also reported to have an increased risk of RCC.³⁰ The data regarding environmental and occupational exposures are inconsistent with the exception of chlorinated solvents.^{23,31}

Moderate alcohol intake,^{32,33} consumption of fruits and (cruciferous) vegetables,^{2,3,34,35} and a diet rich in fatty fish are believed to reduce the risk of RCC. Other studies suggest that non-steroidal anti-inflammatory agents and dietary factors do not play a role in the etiology of RCC.^{5,23,37}

Hereditary and Familial RCC

Family history is associated with an increased risk of RCC and a number of familial RCC syndromes are now well-established, accounting for approximately 4-6% of cases of RCC overall.³⁸ These syndromes include von Hippel-Lindau (VHL), hereditary papillary renal carcinoma (HPRC), Birt Hogg-Dubé (BHD), hereditary leiomyomatosis RCC (HLRCC), succinate dehydrogenase deficiency RCC, tuberous sclerosis, BAP-1 tumor predisposition syndrome, and PTEN hamartoma tumor syndrome (Cowden syndrome). Most of these syndromes have associated tumors or benign findings in other organ systems. RCC in these syndromes tends to be earlier in onset and multifocal and management should prioritize nephron-sparing approaches, including tumor enucleation when feasible to optimize preservation of parenchymal mass. For most of these syndromes, tumors can be observed if less than 3 cm as the risk of metastases remains low in this setting.³⁹ HLRCC and succinate dehydrogenase deficiency RCC are the exception as tumors in these syndromes are often very aggressive and a proactive approach to evaluation and management should be pursued. Genetic counseling should also be strongly recommended for patients suspected of having familial RCC, as it may allow for more intensive evaluation of the patient for RCC and associated manifestations and identification of blood relatives that may be at syndromic risk.

Major Pathological Subtypes

Renal tumors are classified based on cell of origin and morphologic appearance with renal adenocarcinoma being the most common malignant tumor. Major sub-classifications of RCC include clear cell, papillary, chromophobe, collecting duct and unclassified RCC.⁴⁰ A number of uncommon or rare subtypes exist including but not limited to acquired cystic disease-associated RCC, clear cell (tubulo) papillary, and renal medullary carcinoma, which is an aggressive variant typically seen in patients with sickle cell trait. The most common benign tumors of the kidney include oncocytoma and angiomyolipoma (AML). An abbreviated version of the 2016 World Health Organization classification of renal neoplasms is detailed in Table 3.⁴¹

Table 3: Modified 2016 World Health Organization classification of renal neoplasms with focus on adult neoplasms.⁴¹
Renal cell tumors
Clear cell RCC
Multilocular cystic renal neoplasm of low malignant potential
Papillary RCC
HLRCC
Chromophobe RCC
Collecting duct carcinoma
Renal medullary carcinoma
MiT Family translocation carcinomas
Succinate dehydrogenase (SDH) deficient RCC
Mucinous tubular and spindle cell carcinoma
Tubulocystic RCC
Acquired cystic disease associated RCC
Clear cell papillary RCC
RCC, unclassified
Benign renal tumors
Papillary adenoma
Oncocytoma
AML
Metanephric adenoma and other metanephric tumors
Adult cystic nephroma
Mixed epithelial stromal tumors
Juxtaglomerular cell tumor

Table 3: Modified 2016 World Health Organization classification of renal neoplasms with focus on adult neoplasms. ⁴¹
Mescenchymal tumors
Leiomyosarcoma (including renal vein) and other sarcomas
Leiomyoma and other benign mesenchymal tumors
Others
Adults Wilms tumor
Primitive neuroectodermal tumor
Metastatic tumors, lymphoma, leukemia

Presentation and Diagnosis

Presentation

The “classic triad” of symptoms associated with a malignant renal mass include hematuria, flank pain and abdominal mass. Symptoms associated with RCC are often a result of local tumor growth, hemorrhage, paraneoplastic symptoms, or metastatic disease and are uncommon in patients with clinically localized disease. In fact, less than 5 percent of patients in contemporary series present with these symptoms and greater than 50 percent of renal masses are diagnosed incidentally during an evaluation for unrelated signs or symptoms.^{42,43}

Diagnosis

Physical examination has a limited role in the diagnosis of clinically localized disease. However, physical examination may have value in distinguishing the signs and symptoms of advanced disease. For instance, paraneoplastic syndromes (i.e. hypertension, polycythemia, hypercalcemia) are present in approximately 10-20 percent of patients with metastatic RCC.^{6,7,44} Importantly, physical examination of patients with localized disease may occasionally reveal unsuspected adenopathy, varicocele or medical conditions that influence management decisions including body habitus, prior abdominal scars, stigmata of CKD, etc. In addition, careful physical examination may also reveal findings suggestive of familial disease, such as dermatologic lesions.

Laboratory Evaluation

There are no biomarkers or routine laboratory tests used to diagnose renal malignancies. As such, laboratory tests are useful in the assessment of renal function (GFR) and for completeness of metastatic evaluation. Routine laboratory tests for renal mass evaluation include complete metabolic panel, complete blood count, and urinalysis.

Imaging Techniques

Pre and post contrast-enhanced axial imaging, either CT or MRI, is the ideal imaging technique for the diagnosis and staging of clinically localized renal masses. Masses initially diagnosed by US or intravenous pyelography should be confirmed with pre/post contrast-enhanced imaging. Depending on tumor size, 20 to 30 percent of clinically localized renal masses may be benign.^{7,10} Patient and tumor characteristics can indicate populations more or less likely to harbor benign or malignant disease. For instance, women with smaller tumors have a higher likelihood of having benign tumors.^{9,45,46} However, with the exception of fat-containing AML, none of the current imaging modalities can reliably distinguish between benign and malignant tumors or between indolent and aggressive tumor biology.

Contrast-enhanced abdominal imaging (CT or MRI) best characterizes the mass, provides information regarding renal morphology (of the affected and unaffected kidney), assesses extrarenal tumor spread (venous invasion or regional lymphadenopathy) and evaluates the adrenal glands and other abdominal organs for visceral metastases. Based on the most recent consensus statement from the ACR and the National Kidney Foundation, patients with acute kidney injury or CKD and GFR less than 30 mL/min/1.73m² who are not undergoing renal replacement therapy should receive intravenous

normal saline prophylaxis prior to receiving iodinated contrast media.⁴⁷ Patients with GFR of 30-44 mL/min/1.73m² may be considered for intravenous fluid prophylaxis per individual physician discretion based on the patient’s risk factor for renal injury. However, MRI with second generation gadolinium-based intravenous contrast is now a safer option in many patients with severe CKD, as outlined below.

The association of gadolinium-based MRI contrast agents with the development of nephrogenic systemic fibrosis – a devastating and potentially fatal condition has been a concern for many years. More recently, however, with newer group II and III gadolinium-based contrast media the risk is felt to be lower than previously perceived. The most recent consensus statement from the ACR and the National Kidney Foundation suggests that such agents can be given to patients with a GFR under 30 mL/min/1.73m².⁴⁸ A recent systematic review of the risks of NSF in patients with CKD 4 and 5 noted that the risks of NSF using group II gadolinium-based agents was less than 0.07%. Current ACR guidelines on the use of contrast media state that patients need not be screened for renal function prior to receiving group II gadolinium-based agents. Non-contrast CT, MRI (with diffusion weighted images) and US (with Doppler and with or without microbubbles) can also be used to characterize renal masses in patients who cannot receive conventional intravenous contrast.⁴⁷

In general, solid renal masses that enhance greater than 15-20 HU with intravenous contrast and do not exhibit fat density should be considered suspicious for RCC. Approximately 5-10% of AML’s are fat poor and difficult to identify on imaging. Fat poor AML’s often demonstrate suggestive features such as high attenuation on unenhanced CT, homogeneous enhancement on CT, or hypointensity on T2-weighted MR, but the diagnosis remains difficult. Complex cystic renal masses that have thickened irregular walls or septa in which measurable enhancement is present are classified as Bosniak 3. Approximately 50% of such lesions prove to be malignant on final pathology. Bosniak 4 complex cystic lesions are very suspicious for malignancy as they contain enhancing nodular soft tissue components and about 75-90% of such lesions prove to be RCC on final pathology. This guideline focuses primarily on the evaluation and management of clinically localized sporadic renal masses suspicious for RCC in adults, including solid enhancing renal tumors and Bosniak 3 and 4 cystic renal masses.

In patients with RCC or suspicion of RCC, complete staging is typically finalized with chest radiography (CXR) or chest CT. Chest CT scan should be obtained selectively, primarily for patients with pulmonary symptoms or abnormal CXR, or for patients with high-risk disease.^{49,50} Bone scans should be reserved primarily for patients with bone pain or elevated alkaline phosphatase and brain imaging for those with neurologic symptoms.⁵¹⁻⁵³ PET scan has a very limited role in the routine evaluation or staging of RCC.

Renal Mass Biopsy

RMB currently has an adjunctive role in the diagnosis and risk stratification of patients with renal masses suspicious for renal cancer. Biopsy, or FNA, was traditionally reserved for patients suspected of having metastasis of another primary to the kidney, abscess, or lymphoma, or when needed to establish a pathologic diagnosis of RCC in occasional patients presenting with disseminated metastases or unresectable primary tumors. The role of RMB for clinically localized RCC has evolved considerably over the past few decades with substantial variance in practice patterns.

Tumor Characteristics

Staging

Kidney cancer is staged both clinically and pathologically using the staging system outlined by the American Joint Committee on Cancer (AJCC), also known as the tumor node metastases (TNM) classification.⁵⁴ The AJCC TNM Staging System for Kidney Cancer is detailed in Table 4. Stage I and II tumors include cancers of any size that are confined to the kidney. This guideline statement identifies patients with renal masses suspicious for clinical stage I and II RCC, recognizing that a certain number of patients will be upstaged. Stage III tumors are either locally invasive (T3) or have involved lymph nodes (N1). Stage IV tumors have spread beyond the kidney into adjacent organs by direct invasion (T4) or distant metastases (M1). Prognosis is best predicted by stage with cancer-specific survival rates that approximate 85-90% for clinically localized (Stage I and II) RCC.

TABLE 4: THE AJCC TNM STAGING SYSTEM FOR KIDNEY CANCER. ⁴¹

Primary Tumor (T), Regional Lymph Nodes (N) and Distant Metastases (M) are detailed in Table 4A; The Anatomic Stage/Prognostic Groups are detailed in Table 4B.

Table 4A.
Primary Tumor (T)

Table 4A.	
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor.
T1	Tumor ≤7 cm in greatest dimension, limited to the kidney.
T1a	Tumor ≤4 cm in greatest dimension, limited to the kidney.
T1b	Tumor >4 cm but not >7 cm in greatest dimension, limited to the kidney.
T2	Tumor >7 cm in greatest dimension, limited to the kidney.
T2a	Tumor >7 cm but ≤10 cm in greatest dimension, limited to the kidney.
T2b	Tumor >10 cm, limited to the kidney.
T3	Tumor extends into major veins or perinephric tissues but not into the ipsilateral adrenal gland and not beyond Gerota’s fascia.
T3a	Tumor extends into the renal vein or its segmental branches, or invades the pelvicaliceal system, or invades perirenal and/or renal sinus fat but not beyond Gerota’s fascia.
T3b	Tumor grossly extends into the vena cava below the diaphragm.
T3c	Tumor grossly extends into the vena cava above the diaphragm or invades the wall of the vena cava.
T4	Tumor invades beyond Gerota’s fascia (including contiguous extension into the ipsilateral adrenal gland).
Regional Lymph Nodes (N)	
NX	Regional lymph nodes cannot be assessed.
N0	No regional lymph node metastasis.
N1	Metastases in regional lymph node(s).
Distant Metastasis (M)	
M0	No distant metastasis.
M1	Distant metastasis.

Table 4b.			
Stage	T	N	M
I	T1	N0	M0
II	T2	N0	M0
III	T1 or T2	N1	M0
	T3	N0 or N1	M0
IV	T4	Any N	M)
	Any T	Any N	M1

Grading

Historically, a number of grading systems existed and evolved to describe tumor differentiation, cytologic aggressiveness, and prognosis of RCC based on nuclear size and irregularity. In 1982, the Fuhrman Grading system was described and became the most widely used grading system for RCC.⁵⁵ In 2012, the International Society of Urological Pathology (ISUP) Grading System for RCC was proposed and was updated in 2016.^{41,56} The ISUP Grading System incorporates aspects of the Fuhrman Grading system but includes more objective criteria for nuclear characteristics. In addition, sarcomatoid and rhabdoid tumors, tumors with giant cells, and tumors with extreme nuclear pleomorphism are included within grade 4 tumors. Chromophobe RCC is no longer graded in the ISUP system. In general, higher grade is associated with larger tumor size and more aggressive tumors.^{57,58}

Other Prognostic indicators and Nomograms

Other factors for prognostic consideration include tumor size, necrosis, sarcomatoid features, collecting system invasion, patient symptoms, signs of paraneoplastic syndromes, and performance status. Tumor size is important for risk stratification regarding the likelihood of malignancy and more aggressive pathology.^{7,10,46} Various algorithms including the UCLA Integrated Staging System (UISS),^{59,60} Stage, Size, Grade and Necrosis (SSIGN) score,⁶¹⁻⁶³ and other nomograms^{8,9,64} incorporate a variety of pathological and patient characteristics in an effort to improve prognostication.

Other Clinical and Biological Indicators

A number of molecular studies and markers have been proposed for diagnostic and prognostic purposes in RCC. The AHRQ Systematic Review identified a number of biomarkers and laboratory tests that may have diagnostic or prognostic utility in the renal cancer literature.⁶⁵ However, these studies were often univariable in design and therefore excluded from analysis due to a failure to include clinical variables or suboptimal methodology to validate the ultimate value of the tests. Therefore, the AHRQ report identified clinical and biological indicators as a major research gap in the renal cancer literature.⁶⁶

Of note, urine aquaporin-1 and perilipin-2 were identified as emerging biomarkers with potential for the diagnosis of RCC.^{10,11,67,68} Carbonic anhydrase-9 (CAIX) expression is governed by the transcription factor hypoxia-inducible factor-1 α (HIF-1 α), a well-known component of the VHL pathway of clear cell RCC.⁶⁹ While CAIX expression on primary tumors is a prognostic factor, especially in patients with metastatic RCC, high and homogenous levels of CAIX expression prevent risk stratification and clinical utility beyond the established clinical predictors of aggressive, clear cell RCC.⁷⁰ Serum tests including C-reactive protein and platelet count may have prognostic roles, but further investigation is needed. New imaging modalities, including molecular imaging techniques using CAIX⁷¹⁻⁷³ or 99m technetium-sestamibi⁷⁴ single photon emission CT, may help to better differentiate between malignant and benign pathology. However, most markers and imaging modalities in this domain are best characterized as investigational.

Overview of Treatment Alternatives

A number of strategies exist for the management of sporadic renal masses suspicious for clinically localized renal cancer. Four strategies are considered standards of care and include AS, RN, PN, and TA.

Active Surveillance (AS)

A growing body of literature exists regarding AS for patients with clinically localized small renal masses (cT1a, ≤ 4 cm). A number of retrospective studies and meta-analyses evaluate the safety of AS and quote the risk of metastatic progression while on AS to be less than 2 percent in well selected patients over the initial 3 years of AS.⁷⁵⁻⁷⁷ Two large prospective AS programs have been initiated that follow patients with serial imaging, and both report slow growth rates and extremely low rates of metastatic progression, albeit with relatively short follow-up.⁷⁸⁻⁸⁰ Both programs screen patients with an initial metastatic evaluation including serum laboratory evaluation and chest imaging. Patients are then evaluated every 3-6 months for two years and with extended imaging intervals beyond that. Rates of biopsy are variable with one group utilizing RMB in greater than 50 percent of the cohort and the other using biopsy in less than 10 percent of its patients. Further data with longer follow-up from these cohorts will help to inform the utility of AS in the small renal mass population, and should allow for more intelligent patient selection for AS. Of note, the Delayed Intervention and Surveillance for Small Renal Masses (DISSRM) Registry prospectively catalogues a contemporaneous cohort of patients undergoing AS and primary intervention and will offer data regarding comparative effectiveness.⁸⁰

Radical Nephrectomy (RN)

RN was the mainstay of therapy for all renal masses for many decades. Historically, RN included the removal of the entire kidney including Gerota's/Zuckerkandel's fascia, regional lymph nodes and the adrenal gland. RN can be performed through an open incision or via minimally-invasive approaches (laparoscopic or robotic). Cancer-specific survival associated with RN is excellent; however, recent controversies regarding RN include its negative impact on renal function and historical overutilization for the management of stage I, especially T1a, tumors.

Partial Nephrectomy (PN)

PN is widely accepted as a nephron-sparing approach to the management of clinically localized RCC. Initially underutilized and predominantly performed in large academic centers,^{12,13,81} the management of clinically localized renal masses by PN has expanded with implementation of guideline statements and the expansion of robotic technology.^{14,15,82} PN can be performed through an open incision or via a minimally invasive approach, although the robotic approach has largely supplanted laparoscopic surgery as the preferred minimally invasive approach.⁸³ The benefit of PN lies in the potential to preserve renal function but this is counterbalanced by an increased risk of urologic complications, although most are manageable and typically associated with good outcomes. Recent controversies surround modifiable and non-modifiable factors during surgery to improve renal functional outcomes, including parenchymal volume preservation, warm versus cold ischemia, and duration of ischemia.

Thermal Ablation (TA)

TA techniques were developed in an effort to improve patient procedural tolerance and reduce the potential for complications from PN, while still preserving renal function. A multitude of techniques/technologies have been investigated to ablate renal tumors; however, RFA and cryoablation have been most widely investigated and integrated into clinical practice. While the superiority of RFA or cryoablation remains controversial, it is generally accepted that oncologic outcomes are similar for both approaches.⁸⁴⁻⁸⁶ TA has traditionally been performed through a variety of approaches, including open, laparoscopic, and percutaneous. Concerns with the TA literature included relatively limited follow-up, lack of pre and post treatment biopsy to define malignancy and efficacy, and increased local recurrence rates relative to surgical excision. The latter require a longer period of surveillance (5 years) with cross-sectional imaging to monitor for late local recurrences.

Investigational Modalities

Other technologies including high-intensity focused US (HIFU), radiosurgery, microwave therapy, pulsed cavitational US, and laser thermal therapy remain investigational at this time.

Follow-up after Intervention

The prognosis of patients treated with surgery or thermal ablation for kidney cancer is primarily determined by tumor stage, with tumor size, grade, histology, with a variety of other contributing factors.^{61,62,87-93} Several algorithms and prognostic models have been published, yet a recent analysis of outcomes from a phase III randomized adjuvant clinical trial suggested that these models only marginally outperformed stage alone.⁹⁴

Current surveillance and survivorship strategies for patients with RCC have incorporated clinical history, physical examination, relevant laboratory testing, and abdominal and chest imaging.^{181,387,95} This allows for assessment of potential complications or sequelae of intervention, functional recovery, and evaluation for common sites of recurrence, including those in the lungs, liver, adrenal glands, and other retroperitoneal sites. Cross-sectional imaging is generally preferred, particularly for high-risk patients, while abdominal ultrasound or chest radiography can be considered in lower-risk patients or as a potential alternative during long-term surveillance. Approximately thirty-percent of recurrences have been diagnosed after 5 years in some series, emphasizing the need to consider longer follow-up than advocated in most current surveillance protocols.¹ Bone metastasis are only rarely identified during surveillance in the absence of bone pain, an elevated alkaline phosphatase, or radiographic findings suggesting a bony neoplasm, and bone scan can generally be reserved for these indications.⁹⁶⁻⁹⁹ Patients with acute neurological signs/symptoms should undergo prompt cross-sectional imaging of the brain and/or spine,³⁷⁰⁻³⁷² but beyond this there is no role for routine neurologic imaging in surveillance of patients with localized renal cancer. Additional site-specific imaging should be ordered as warranted by clinical signs/symptoms suggestive of recurrence or metastatic spread. Current data do not support the use of PET scan in the routine surveillance of patients with renal cancer, and this test should only be considered selectively, such as for troubleshooting when other tests are concerning but inconclusive.³⁷³

There are no prospective data to compare currently available surveillance strategies, resulting in substantial variability in the approach, modality, frequency, and duration of follow-up after intervention. The premise of early detection of tumor recurrence after primary intervention is that this approach will result in patient cure, improved survival, or appropriate palliation. In addition, surveillance allows the urologist to provide a measure of reassurance to the patient who is worried about cancer recurrence. Surveillance also offers the opportunity to monitor treatment effects and address survivorship issues that might arise. Taking all of these considerations into account, the Panel updated the follow-up strategies after intervention to strike a useful and measured balance.

Evaluation and Diagnosis

EVALUATION

GUIDELINE STATEMENT 1

1. In patients with a solid or complex cystic renal mass, clinicians should obtain high quality, multiphase, cross-sectional abdominal imaging to optimally characterize and clinically stage the renal mass. Characterization of the renal mass should include assessment of tumor complexity, degree of contrast enhancement (where applicable), and presence or absence of fat. (Clinical Principle)

Discussion

GUIDELINE STATEMENT 2

2. In patients with suspected renal malignancy, clinicians should obtain a comprehensive metabolic panel, complete blood count, and urinalysis. Metastatic evaluation should include chest imaging to evaluate for possible thoracic metastases. (Clinical Principle)

Discussion

GUIDELINE STATEMENT 3

3. For patients with a solid or Bosniak 3/4 complex cystic renal mass, clinicians should assign chronic kidney disease (CKD) stage based on glomerular filtration rate (GFR) and degree of proteinuria. (Expert Opinion)

Discussion

Counseling

GUIDELINE STATEMENT 4

In patients with a solid or Bosniak 3/4 complex cystic renal mass, a urologist should lead the counseling process and should consider all management strategies. A multidisciplinary team should be included when necessary. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 5

Clinicians should provide counseling that includes current perspectives about tumor biology and a patient-specific risk assessment inclusive of sex, tumor size/complexity, histology (when obtained), and imaging characteristics. For cT1a tumors, the low oncologic risk of many small renal masses should be reviewed. (Clinical Principle)

Discussion

GUIDELINE STATEMENT 6

During counseling of patients with a solid or Bosniak 3/4 complex cystic renal mass, clinicians must review the most common and serious urologic and non-urologic morbidities of each treatment pathway and the importance of patient age, comorbidities/frailty, and life expectancy. (Clinical Principle)

Discussion

GUIDELINE STATEMENT 7

Clinicians should review the importance of renal functional recovery related to renal mass management, including the risks of progressive CKD, potential short- or long-term need for renal replacement therapy, and long-term overall survival considerations. (Clinical Principle)

Discussion

GUIDELINE STATEMENT 8

Clinicians should consider referral to nephrology in patients with a high risk of CKD progression, including those with estimated glomerular filtration rate (eGFR) less than 45 mL/min/1.73m², confirmed proteinuria, diabetics with preexisting CKD, or whenever eGFR is expected to be less than 30 mL/min/1.73m² after intervention. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 9

Clinicians should recommend genetic counseling for any of the following: all patients ≤ 46 years of age with renal malignancy, those with multifocal or bilateral renal masses, or whenever 1) the personal or family history suggests a familial renal neoplastic syndrome; 2) there is a first-or second-degree relative with a history of renal malignancy or a known clinical or genetic diagnosis of a familial renal neoplastic syndrome (even if kidney cancer has not been observed); or 3) the patient’s pathology demonstrates histologic findings suggestive of such a syndrome. (Expert Opinion)

Discussion

Renal Mass Biopsy (RMB)

GUIDELINE STATEMENT 10

When considering the utility of RMB, patients should be counseled regarding rationale, positive and negative predictive values, potential risks and non-diagnostic rates of RMB. (Moderate Recommendation; Evidence Level: Grade C)

Discussion

GUIDELINE STATEMENT 11

Clinicians should consider RMB when a mass is suspected to be hematologic, metastatic, inflammatory, or infectious. (Clinical Principle)

Discussion

GUIDELINE STATEMENT 12

In the setting of a solid renal mass, RMB should be obtained on a utility-based approach whenever it may influence management. RMB is not required for 1) young or healthy patients who are unwilling to accept the uncertainties associated with RMB; or 2) older or frail patients who will be managed conservatively independent of RMB findings. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 13

For patients with a solid renal mass who elect RMB, multiple core biopsies should be performed and are preferred over fine needle aspiration (FNA). (Moderate Recommendation; Evidence Level: Grade C)

Discussion

Management

PARTIAL NEPHRECTOMY (PN) AND NEPHRON-SPARING APPROACHES

Guideline Statement 14

Clinicians should prioritize PN for the management of the cT1a renal mass when intervention is indicated. In this setting, PN minimizes the risk of CKD or CKD progression and is associated with favorable oncologic outcomes, including excellent local control. (Moderate Recommendation; Evidence Level: Grade B)

Discussion

GUIDELINE STATEMENT 15

Clinicians should prioritize nephron-sparing approaches for patients with solid or Bosniak 3/4 complex cystic renal masses and an anatomic or functionally solitary kidney, bilateral tumors, known familial RCC, preexisting CKD, or proteinuria. (Moderate Recommendation; Evidence Level: Grade C)

Discussion

GUIDELINE STATEMENT 16

Nephron-sparing approaches should be considered for patients with solid or Bosniak 3/4 complex cystic renal masses who are young, have multifocal masses, or comorbidities that are likely to impact renal function in the future, including but not limited to moderate to severe hypertension, diabetes mellitus, recurrent urolithiasis, or morbid obesity. (Moderate Recommendation; Evidence Level: Grade C)

Discussion

GUIDELINE STATEMENT 17

In patients who elect PN, clinicians should prioritize preservation of renal function by optimizing nephron mass preservation and avoiding prolonged warm ischemia. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 18

For patients undergoing PN, clinicians should prioritize negative surgical margins. The extent of normal parenchyma removed should be determined by surgeon discretion taking into account the clinical situation and tumor characteristics, including growth pattern, and interface with normal tissue. Tumor enucleation should be considered in patients with familial RCC, multifocal disease, or severe CKD to optimize parenchymal mass preservation. (Expert Opinion)

Discussion

RADICAL NEPHRECTOMY (RN)

Guideline Statement 19

Clinicians should consider RN for patients with a solid or Bosniak 3/4 complex cystic renal mass whenever increased oncologic potential is suggested by tumor size, RMB (if obtained), and/or imaging. (Moderate Recommendation; Evidence Level: Grade B) In this setting, RN is preferred if all of the following criteria are met: 1) high tumor complexity and PN would be challenging even in experienced hands; 2) no preexisting CKD or proteinuria; and 3) normal contralateral kidney and new baseline eGFR will likely be greater than 45 mL/min/1.73m2 even if RN is performed. If all of these criteria are not met, PN should be considered unless there are overriding concerns about the safety or oncologic efficacy of PN. (Expert Opinion)

Discussion

Surgical Principles

GUIDELINE STATEMENT 20

For patients who are undergoing surgical excision of a renal mass with clinically concerning regional lymphadenopathy, clinicians should perform a lymph node dissection including all clinically positive nodes for staging purposes. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 21

For patients who are undergoing surgical excision of a renal mass, clinicians should perform adrenalectomy if imaging and/or intraoperative findings suggest metastasis or direct invasion of the adrenal gland. (Clinical Principle)

Discussion

GUIDELINE STATEMENT 22

In patients undergoing surgical excision of a renal mass, a minimally invasive approach should be considered when it would not compromise oncologic, functional, and perioperative outcomes. (Expert Opinion)

Discussion

Other Considerations

GUIDELINE STATEMENT 23

Pathologic evaluation of the adjacent renal parenchyma should be performed and recorded after PN or RN to assess for possible intrinsic renal disease, particularly for patients with CKD or risk factors for developing CKD. (Clinical Principle)

Discussion

GUIDELINE STATEMENT 24

Clinicians should consider referral to medical oncology whenever there is concern for potential clinical metastasis or incompletely resected disease (macroscopic positive margin or gross residual disease). Patients with high-risk or locally advanced, fully resected renal cancers should be counselled about the risks/benefits of adjuvant therapy and encouraged to participate in adjuvant clinical trials, facilitated by medical oncology consultation when needed. (Clinical Principle)

Discussion

Thermal Ablation (TA)

GUIDELINE STATEMENT 25

Clinicians should consider TA as an alternate approach for the management of cT1a solid renal masses <3 cm in size. For patients who elect TA, a percutaneous technique is preferred over a surgical approach whenever feasible to minimize morbidity. (Moderate Recommendation; Evidence Level: Grade C)

Discussion

GUIDELINE STATEMENT 26

Both radiofrequency ablation (RFA) and cryoablation may be offered as options for patients who elect TA. (Conditional Recommendation; Evidence Level: Grade C)

Discussion

GUIDELINE STATEMENT 27

A RMB should be performed prior to (preferred) or at the time of ablation to provide pathologic diagnosis and guide subsequent surveillance. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 28

Counseling about TA should include information regarding an increased likelihood of tumor persistence or local recurrence after primary TA relative to surgical excision, which may be addressed with repeat ablation if further intervention is elected. (Strong Recommendation; Evidence Level: Grade B)

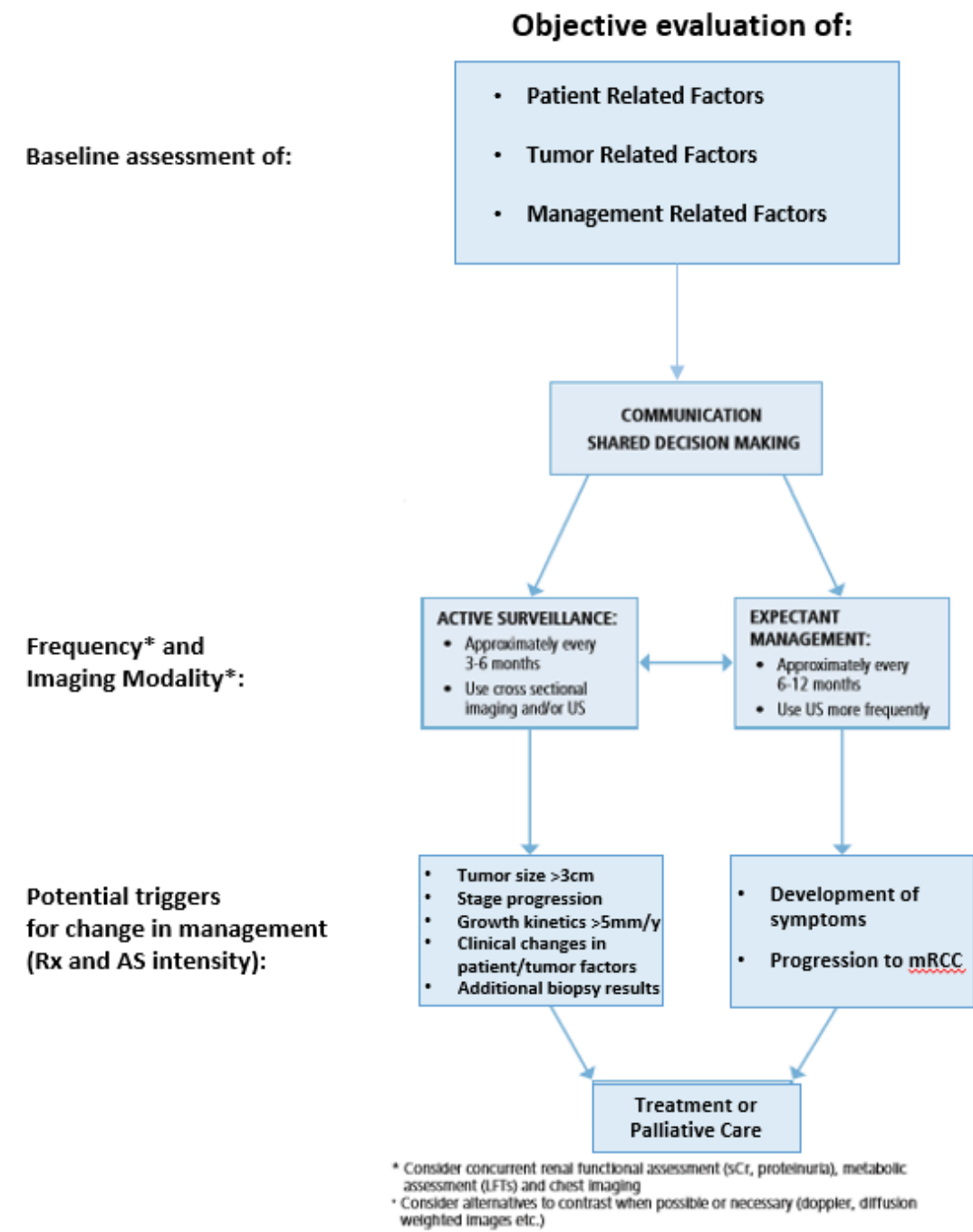
Discussion

Active Surveillance

Introduction

The decision to embark on a course of AS or expectant management rather than treatment in the setting of a localized renal mass presumed to be a renal cancer requires thoughtful consideration by both the patient and the physician. In making the decision, an objective baseline evaluation of patient, tumor, and treatment-related factors should be undertaken (Figure 7). This should include formal decision-making tools whenever possible leading to a well communicated risk-benefit analysis unique to the individual patient’s circumstances.^{110,344-347} The shared decision-making process should be consistent with the patient’s inherent preferences and tolerance of uncertainty.³⁴⁸

FIGURE 7. ALGORITHM FOR AS OR EXPECTANT MANAGEMENT OF LOCALIZED RENAL MASSES SUSPICIOUS FOR MALIGNANCY.



High level data regarding the optimal frequency and preferred imaging modalities for renal mass surveillance are lacking. Therefore, at the time of the initial baseline assessment and during subsequent re-assessments, the clinician should estimate how to best achieve the goals of (1) preventing stage progression, (2) maintaining renal function and (3) avoiding the potential risks of treatment when it is unlikely to provide an oncologic or survival benefit. At the onset of AS, the clinician should request and evaluate prior abdominal imaging that may demonstrate the existence of the renal mass at an earlier time point to assess growth rate or changes in clinical stage. Next, patients placed on a program of non-intervention should be considered for either AS or expectant management (observation or watchful waiting) (Figure 7).

AS is most appropriate for patients in whom the anticipated net benefit of AS is modest to significant when compared to treatment. Excluded from this track are patients who are reasonable candidates for intervention if tumor size, infiltrative appearance, interval growth, or RMB suggest the potential for cancer progression, unless they are willing to accept the associated increase in oncologic risk (see statement 31 and 32 below). Patients with no prior imaging should have surveillance imaging initially every 3 to 6 months to assess for interval growth, substantial radiographic changes in the character of the lesion, or the presence of rare occult synchronous metastases in the setting of a small renal mass. The preferred modality is not well established in the literature, but initial imaging should preferably consist of contrast-enhanced cross-sectional imaging. Subsequent imaging may include the same or when appropriate an abdominal US can be substituted. Abdominal US (as opposed to retroperitoneal US), may have the additional benefit of a survey of the intraabdominal organs for progression. Differences in tumor dimension measurements between these different modalities

may be significant and should be interpreted with caution when making treatment decisions.⁸⁰ RMB can be considered for additional risk stratification for patients with solid masses on AS. For those with predominantly cystic lesions, RMB should be avoided.

It is recognized that not all patients on AS will require the same intensity of surveillance as their tumor biology, risk calculations and tradeoffs, and personal objectives may differ. Some patients may therefore require more intensive AS while others require less intensive AS. The decision as to the frequency and imaging modality must therefore be customized and informed by robust communication focusing on goals, risks and triggers for intervention. RMB can be a helpful adjunct to guide these clinical decisions (see statement 10). However, even when RMB suggests the tumor is benign, the predictive value of a core biopsy is imperfect due to tumor heterogeneity and the possibility of collision tumors.^{192,193} Currently there are insufficient data to recommend that all patients with benign RMBs can be advised that they no longer need follow-up imaging. Judicious surveillance in appropriate patients with benign appearing RMBs remains a prudent strategy.

Expectant management (observation) is appropriate in patients in whom treatment poses an unacceptably high periprocedural or renal functional risk than surveillance. In this setting, the use of abdominal US of the retroperitoneal and intraperitoneal organs can be performed more frequently than formal contrast based cross-sectional imaging to screen for stage progression which may trigger systemic or palliative therapy in the appropriately selected patient.

Regardless of the intensity of surveillance, chest imaging with plain radiography (CXR) is warranted annually or if intervention triggers are encountered or symptoms arise. The intensity of surveillance can be attenuated if the renal mass exhibits slow growth kinetics, is noted to be radiographically stable or if the patient’s medical condition deteriorates. In cases such as this, patients can cross over between AS and expected management (observation) based on changing risk profiles, performance status, absolute tumor size, tumor growth kinetics, stage progression or other recalibration triggers for possible intervention.^{349,350} While no level 1 data exist that define these triggers precisely, they should generally be based on changes in tumor-based risk (absolute size > 3cm, median growth rate in excess of 5mm/year, or stage migration) or patient-based risks (co-morbidities) with continual objective reassessments to include the use of RMB when appropriate.^{349,350} Published data demonstrate that in most instances, judicious delayed intervention for localized stage I renal masses remains effective.^{159,349-354}

The key to successful AS of a localized renal mass remains thoughtful and recurrent reassessments and robust communication in partnership with the patient and his/her caregivers. Prospective trials, ideally randomized, of AS versus treatment, with improved reporting and more extended follow-up, should be prioritized to provide higher quality data about oncologic, functional and survival outcomes.

GUIDELINE STATEMENT 29

For patients with a solid renal mass < 2cm, or those that are complex but predominantly cystic, clinicians may elect AS with potential for delayed intervention for initial management. (Conditional Recommendation; Evidence Level: Grade C)

Discussion

GUIDELINE STATEMENT 30

For patients with a solid or Bosniak 3/4 complex cystic renal mass, clinicians should prioritize AS/expectant management when the anticipated risk of intervention or competing risks of death outweigh the potential oncologic benefits of active treatment. In asymptomatic patients, the panel recommends periodic clinical surveillance and/or imaging based on shared decision making. (Clinical Principle)

Discussion

GUIDELINE STATEMENT 31

For patients with a solid or Bosniak 3/4 complex cystic renal mass in whom the risk/benefit analysis for treatment is equivocal and who prefer AS, clinicians should consider RMB (if the mass is solid or has solid components) for further oncologic risk stratification. Repeat cross-sectional imaging should be obtained approximately 3-6 months later to assess for interval growth. Periodic clinical/imaging surveillance can then be based on growth rate and shared decision-making with intervention recommended if substantial interval growth is observed or if other clinical/imaging findings suggest that the risk/benefit analysis is no longer equivocal or favorable for continued AS. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 32

For patients with a solid or Bosniak 3/4 complex cystic renal mass in whom the anticipated oncologic benefits of intervention outweigh the risks of treatment and competing risks of death, clinicians should recommend intervention. AS with potential for delayed intervention may be pursued only if the patient understands and is willing to accept the associated oncologic risks. In this setting, clinicians should encourage RMB (if the mass is predominantly solid) for additional risk stratification. If the patient continues to prefer AS, close clinical and cross-sectional imaging surveillance with periodic reassessment and counseling should be recommended. (Moderate Recommendation; Evidence Level: Grade C)

Discussion

Follow-Up After Intervention

GENERAL PRINCIPLES

Guideline Statement 33

Clinicians coordinating follow-up for patients who have undergone intervention for a renal mass should discuss the implications of stage, grade, and histology including the risks of recurrence and possible sequelae of treatment. Patients with pathologically-proven benign renal masses should undergo occasional clinical evaluation and laboratory testing for sequelae of treatment but most do not require routine periodic imaging. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 34

Patients with treated malignant renal masses should undergo periodic medical history, physical examination, laboratory studies, and imaging directed at detecting signs and symptoms of metastatic spread and/or local recurrence as well as evaluation for possible sequelae of treatment. (Clinical Principle)

Discussion

GUIDELINE STATEMENT 35

Patients with treated malignant renal masses should have periodic laboratory testing including serum creatinine, eGFR, and urinalysis. Other laboratory evaluations (e.g., complete blood count, lactate dehydrogenase, liver function tests, alkaline phosphatase and calcium level) may be obtained at the discretion of the clinician or if advanced disease is suspected. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 36

Patients undergoing follow-up for treated renal masses with progressive renal insufficiency or proteinuria should be referred to nephrology. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 37

Patients undergoing follow-up for treated malignant renal masses should only undergo bone scan if one or more of the following is present: clinical symptoms such as bone pain, elevated alkaline phosphatase, or radiographic findings suggestive of a bony neoplasm. (Moderate Recommendation; Evidence Level: Grade C)

Discussion

GUIDELINE STATEMENT 38

Patients undergoing follow-up for treated malignant renal masses with acute neurological signs or symptoms should undergo prompt magnetic resonance imaging (MRI) or computed tomography (CT) scanning of the brain and/or spine. (Strong Recommendation; Evidence Level: Grade A)

Discussion

GUIDELINE STATEMENT 39

For patients undergoing follow-up for treated malignant renal masses, additional site-specific imaging can be ordered as warranted by clinical symptoms suggestive of recurrence or metastatic spread. Positron emission tomography (PET) scan should not be obtained routinely but may be considered selectively. (Moderate Recommendation; Evidence Level: Grade C)

Discussion

GUIDELINE STATEMENT 40

Patients with findings suggestive of metastatic renal malignancy should be evaluated to define the extent of disease and referred to medical oncology. Surgical resection or ablative therapies should be considered in select patients with isolated or oligo-metastatic disease. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 41

Patients with findings suggesting a new renal primary or local recurrence of renal malignancy should undergo metastatic evaluation including chest and abdominal imaging. If the new primary or recurrence is isolated to the ipsilateral kidney and/or retroperitoneum, a urologist should be involved in the decision-making process, and surgical resection or ablative therapies may be considered. (Expert Opinion)

Discussion

Follow-up after Surgery

GUIDELINE STATEMENT 42

Clinicians should classify patients who have been managed with surgery (PN or RN) for a malignant renal mass into one of the following risk groups for follow-up:

Low Risk (LR)	pT1 and Grade 1/2
Intermediate Risk (IR)	pT1 and Grade 3/4, or pT2 any Grade
High Risk (HR)	pT3 any Grade
Very High Risk (VHR)	pT4 or pN1, or sarcomatoid/rhabdoid dedifferentiation, or macroscopic positive margin

If final microscopic surgical margins are positive for cancer, the risk category should be considered at least one level higher, and increased clinical vigilance should be exercised. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 43

Patients managed with surgery (PN or RN) for a renal malignancy should undergo abdominal imaging according to Table 1, with CT or MRI pre- and post-intravenous contrast preferred. (Moderate Recommendation; Evidence Strength: Grade C). After 2 years, abdominal ultrasound (US) alternating with cross-sectional imaging may be considered in the LR and IR groups at physician discretion. After 5 years, informed/shared decision-making should dictate further abdominal imaging. (Expert Opinion)

Discussion

GUIDELINE STATEMENT 44

Patients managed with surgery (PN or RN) for a renal malignancy should undergo chest imaging (chest x-ray [CXR] for LR and IR; CT chest preferred for HR and VHR) according to Table 1. (Moderate Recommendation; Evidence Strength: Grade C). After 5 years, informed/shared decision-making discussion should dictate further chest imaging and CXR may be utilized instead of chest CT for HR and VHR (Expert Opinion)

Discussion

Follow-up after TA

GUIDELINE STATEMENT 45

Patients undergoing ablative procedures with biopsy that confirmed malignancy or was non-diagnostic should undergo pre- and post-contrast cross-sectional abdominal imaging within 6 months (if not contraindicated). Subsequent follow-up should be according to the recommendations for the IR postoperative protocol (Table 1). (Expert Opinion)

Discussion

Future Directions

The most promising routes to advance the field in localized renal cancer include (1) clinical trials, (2) collaborative quality initiatives, (3) novel diagnostics/biomarkers, and (4) improved technologies and systemic therapies. Each of these requires an unrelenting commitment to continuous clinical improvement and scientific investigation.

The management of localized renal cancer is an area for which there is a paucity of randomized clinical trials (RCT’s). Improving the strength of evidence will require an increased commitment to clinical trial design, conduct, and funding. Although our understanding of the nature and management of this disease continues to progress, without adequate engagement and support, our treatment paradigms will likely continue to be more art than science.

An appropriate companion to RCT’s is the development of collaborative quality initiatives (CQI’s).³⁹¹ Within a CQI, participating hospitals and providers collect, share, and analyze data through clinical registries. CQI participants design and affect changes that improve outcomes of complex, highly technical areas of care.³⁹² CQI registries allow for a more robust analysis of the link between processes and outcomes than can occur with retrospective single or multi-institutional studies; particularly as more sensitive and specific diagnostics/biomarkers are complemented by technologic advances. Scientific inquiry will continue to provide fundamental knowledge regarding the biological basis, inherent risks, and natural history of localized renal masses such that appropriate trade-offs can be made when considering optimal management.

Evaluation and Diagnosis

The localized renal mass remains primarily a radiographic diagnosis. The field of tumor radiomics, artificial intelligence and molecular imaging promises³⁹³ to improve our ability to discriminate tumor histology, grade^{394,395} and ultimately gene and protein expression with prognostic implications. The development of more sophisticated modeling of patient demographic features as recorded in the electronic medical record, such as age, gender, race, body mass index, comorbidities, exposure to

tobacco, and other risk factors are being studied to contextualize and individualize management options. Finally, tumor markers detected in biopsy, blood, or urine should be studied to improve prognostic models for RCC. Efforts based on gene and protein expression have identified multiple promising markers that may one day distinguish between subtypes of malignant and benign renal tumors.^{396,397} Recent work through The Cancer Genome Atlas (TCGA)^{398,399} to identify genomic markers for clear cell RCC⁴⁰⁰ papillary RCC⁴⁰¹ and chromophobe RCC⁴⁰² holds great clinical potential for more accurate diagnosis, prognostication, and surveillance of renal masses. The promise of measuring circulating tumor cells, or liquid tumor biopsies, for diagnosis and surveillance for recurrence and response to treatment is likely several years off, but could substantially transform care models.⁴⁰³⁻⁴⁰⁶

Counseling and Outcomes-based Research

As data emerge regarding variability in treatments performed for localized renal cancer, the impact of the individual physician-patient interaction becomes more evident. The quality of patient counseling can only be improved by providing high quality data, particularly from RCT's. Given our current state of knowledge, translation of information from research studies and guidelines into practical materials for patients is not straight-forward. The development of decision aids for informed medical-decision making is ongoing.^{407,408} The appropriate application of data from large registries and implementation sciences to improve processes and standardization of care is an important initiative that must move forward. Increased quality of data, including improved assessment of tumor biology and prospective trials of management options, is greatly needed to facilitate more intelligent patient counseling.

Management

A major limitation of the literature supporting the current guidelines for management of localized renal cancer is the relatively low level of evidence. Prospective comparative trials, ideally randomized, comparing AS vs. active intervention (TA or excision) should be prioritized to provide higher quality data about oncologic and renal functional outcomes and to assess the treatment-related morbidities or limitations of each approach. With improved reporting and more extended follow-up, multi-institutional observational data will strengthen confidence in recommendations, but not nearly to the extent that clinical trials can provide.

Comparison of extirpative treatment modalities should include prospective evaluation of PN versus RN, prioritized in patients with a normal contralateral kidney and no preexisting CKD/albuminuria, with the goal of assessing the impact of new baseline functional status on overall survival, cardiovascular health, and subsequent renal stability on a longitudinal basis. Ideally, patients with tumors with increased oncologic potential (cT1b/T2) should be prioritized for such trials.^{259,409,410} Regarding nephron-sparing surgery, improved data comparing the relative merits and limitations of standard PN versus tumor enucleation should be sought, ideally through prospective evaluation incorporating improved reporting, and standard assessment of surgical margins.²⁴⁸

Multiple non-extirpative methods being actively investigated in the management of renal masses include stereotactic body radiation therapy (SBRT), HIFU, microwave ablation (MWA), and laser interstitial thermal therapy (LITT). These approaches differ in their mechanisms of action, invasiveness, reported outcomes and experience. Their use should be approached systematically and with caution, and they should be considered investigational at present. SBRT, also frequently referred to as stereotactic ablative radiotherapy (SABR), has been reported in a small number of series. SBRT involves relatively intense protocols (24 to 40 Gy) over one to five fractions and a high degree of spatial precision, offering the potential to be less invasive than surgical or conventional ablative approaches.⁴¹¹ Despite encouraging results, the current body of evidence is limited due to small patient numbers, short follow-up and inconsistent methods of reporting outcomes.⁴¹¹ Thus, SBRT in the management of localized renal masses at present remains investigational and should be primarily considered for patients who are medically inoperable and are not candidates for established TA approaches. Investigation through clinical trials should be prioritized.

Similarly, HIFU remains investigational in the management of renal masses, although it is currently used clinically to treat prostate cancer and uterine fibroids.⁴¹² HIFU relies on the use of a lens or focused transducer to deliver high-frequency sound waves to tissue, typically 1 to 5 MHz. HIFU may be administered in an entirely noninvasive means similar to extracorporeal lithotripsy, thus minimizing the risk of tumor seeding, urinary extravasation or hemorrhage.⁴¹³ Initial clinical investigations have established the feasibility of transcutaneous HIFU; however, distinct regions of renal masses are frequently left untreated resulting in incomplete ablation.⁴¹⁴⁻⁴¹⁷

Similar to RFA, MWA delivers electromagnetic energy through flexible probes inserted into a target lesion. MWA produces target temperatures (>60° C) more rapidly than RFA, and, thus, appears to have significant potential as an ablative modality.⁴¹⁸ LITT uses optical fibers that are inserted directly into the target tissue to deliver laser light that is converted into thermal energy. The most common laser type used in LITT is a neodymium: yttrium-aluminum-garnet (Nd:YAG) laser.⁴¹⁹ Outcomes of clinical investigations are limited due to the small number of treated patients and short follow-up.^{420,421} Given

the limited number of published studies involving HIFU, MWA and LITT and lack of long-term follow-up, appropriate use of these modalities in the management of small renal masses remains poorly defined. Larger prospective trials will be necessary to develop and assess optimal use, risks and morbidity.

Follow-up after Intervention

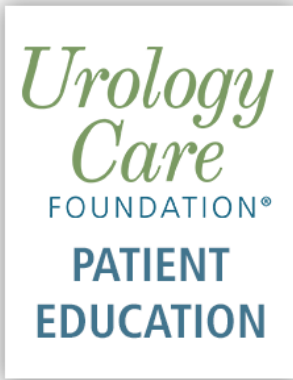
The proposed guidelines for follow-up after intervention for renal cancer attempt to provide a risk-based approach to surveillance and monitoring. Few high-quality studies currently exist to help formulate surveillance regimens, and many of the Panel’s recommendations are thus based primarily on expert opinion. Any cancer surveillance regimen is a balancing act that includes many variables such as the likelihood of disease recurrence at various sites, temporal considerations, the potential benefits of therapeutic interventions and effectiveness of these modalities based on timing of recurrence detection, improvements in diagnostic and initial interventions, patient characteristics, and the burden and cost of monitoring. As electronic medical records and quality and safety initiatives intensify, tracking outcomes of all patients will become increasingly codified and more usable for research purposes. These data can then also be used to inform the proper sequencing, timing, duration, and type of follow-up that improves patient outcomes with the most parsimonious monitoring.

Future research to make patient follow-up more efficient and effective could include one or many of these modalities: develop circulating biomarkers to supplement currently available imaging, develop novel functional imaging, conduct clinical trials to compare currently available imaging modalities, as well as clinical trials to guide the *frequency* of imaging/follow-up, similar to studies done in testicular cancer (MRC TE08)⁴²², colon cancer (GILDA)⁴²³, and non-small cell lung cancer (IFCT-0302).⁴²⁴

Summary

In conclusion, improving the management of localized renal tumors will require a concerted effort among clinicians and allied fields to develop higher quality evidence and facilitate more precise estimations of relative risks and benefits of each therapeutic approach.

Tools and Resources



[Kidney Cancer Filtered Resources](#)

Abbreviations

Active Surveillance	AS
Agency for Healthcare Reserach and Quality	AHRQ
American College of Radiology	ACR
American Society of Nephrology	ASN
American Urological Association	AUA
Angiomyolipoma	AML
Birt Hogg-Dube	BHD
Chest X-Ray	CXR
Chronic Kidney Disease	CDK
College of American Pathologists	CAP
Computed Tomography	CT
End-Stage Renal Disease	ESRD
Estimated Glomerular filtration rate	eGFR

Fine Needle Aspiration	FNA
Glomerular Filtration Rate	GFR
Hereditary Leiomyomatosis RCC	HLRCC
Hereditary Papillary Renal Carcinoma	HPRC
High Risk	HR
High-Intensity Focused Ultrasound	HIFU
Intermediate Risk	IR
Laser Interstitial Thermal Therapy	LITT
Low Risk	LR
Magnetic Resonance Imaging	MRI
Microwave Ablation	MWA
Partial Nephrectomy	PN
Positron Emission Tomography	PET
Practice Guidelines Committee	PGC
Radical Nephrectomy	RN

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